THE COHOMOLOGY SHEAVES OF THE DERIVED PULLBACK OF A QUASI-COHERENT MODULE ON THE BLOWUP ARE ANNIHILATED BY A FIXED POWER OF THE IDEAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The affine open covering of the blowup by the spectra of the affine blowup algebras A[I/a_i]. Invoked inside the counted proof at source lines 4651-4655 as a cross-chapter pointer. Reduces the statement about sheaves on the blowup to statements about the affine blowup algebras.
- The description of the derived pullback of the quasi-coherent module on the affine pieces, and the annihilation statement for the affine blowup algebra. The derived pullback description is a cross-chapter pointer (source lines 4656-4658), and the annihilation for the affine case is the author lemma lemma-annihilates-affine, whose statement is reproduced verbatim as a context span of this card (source lines 4554-4560). Turns the required annihilation of the cohomology sheaves into the annihilation of the Tor modules of the affine blowup algebras.

Source reference. Original reference divisors-lemma-blowing-up-affine: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 4655–4655. Navigation pointer only: the counted text refers to the Stacks item divisors-lemma-blowing-up-affine, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference perfect-lemma-quasi-coherence-pullback: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 4658–4658. Navigation pointer only: the counted text refers to the Stacks item perfect-lemma-quasi-coherence-pullback, which belongs to another chapter of the project and is not

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Source reference. Original reference coherent-lemma-coherent-on-proj: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 4612–4612. Navigation pointer only: the counted text refers to the Stacks item coherent-lemma-coherent-on-proj, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference coherent-lemma-recover-tail-graded-module: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 4613–4613. Navigation pointer only: the counted text refers to the Stacks item coherent-lemma-recover-tail-graded-module, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *In the situation above, let t be an upper bound on the number of generators for I. There exists an integer $c = c(A, I) \geq 0$ such that for any A-module M the cohomology sheaves $H^j(Lp^*\widetilde{M})$ are annihilated by I^c for $j \leq -t$.*

For the rest of the discussion in this section we fix a Noetherian ring A and an ideal $I \subset A$. We denote

$$p : X \to \mathrm{Spec}(A)$$

the blowing up of $\mathrm{Spec}(A)$ in the ideal I. In other words, X is the Proj of the Rees algebra $\bigoplus_{n \geq 0} I^n$. By Cohomology of Schemes, Lemmas coherent-lemma-coherent-on-proj and coherent-lemma-recover-tail-graded-module we can choose an integer $q(A, I) \geq 0$ such that for all $q \geq q(A, I)$ we have $H^i(X, \mathcal{O}_X(q)) = 0$ for $i > 0$ and $H^0(X, \mathcal{O}_X(q)) = I^q$.

Lemma 0.2. *Let $I = (a_1, \ldots, a_t)$ be an ideal of a Noetherian ring A. Set $a = a_1$ and denote $B = A[\frac{I}{a}]$ the affine blowup algebra. There exists a $c > 0$ such that $Tor_i^A(B, M)$ is annihilated by I^c for all A-modules M and $i \geq t$.*

Proof. Say $I = (a_1, \ldots, a_t)$. The question is affine local on X. For $1 \leq i \leq t$ let $B_i = A[\frac{I}{a_i}]$ be the affine blowup algebra. Then X has an affine open covering by the spectra of the rings B_i, see Divisors, Lemma divisors-lemma-blowing-up-affine. By the description of derived pullback given in Derived Categories of Schemes, Lemma perfect-lemma-quasi-coherence-pullback we conclude it suffices to prove that for each i there exists a $c \geq 0$ such that

$$\mathrm{Tor}_j^A(B_i, M)$$

is annihilated by I^c for $j \geq t$. This is Lemma 0.2. □